A Novel Autofocus Framework for UAV SAR Imagery: Motion Error Extraction From Symmetric Triangular FMCW Differential Signal

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***Abstract*— Synthetic aperture radar (SAR) installed on unmanned aerial vehicles (UAVs) has drawn increasing attention in the area of remote sensing due to its high spatial resolution and low energy consumption. However, due to the sensitivity toward atmospheric turbulences, strong motion errors can defocus the image in both azimuth and range directions. In this article, based on symmetric triangular linear frequency-modulation continuous microwave (STLFMCW) signals, a novel motion error estimation method is proposed. The core concept is to eliminate the effect of residual range cell migration (RCM) and investigate the motion parameters from azimuth phase history by range-frequency-domain interferometry between the up-ramp and down-ramp chirp sections. With preknowledge of the structural characteristics of the differential signal, we first obtain the high-order component, then estimate the quadratic term with an azimuth profile width minimization (APWM) algorithm on the basis of bisection search, and subsequently remove the linear part through phase gradient matching and joining. Consequently, the desired motion errors can be achieved purely based on raw data. Simulations demonstrate that our proposed algorithm can achieve high accuracy with or without standard prominent point targets. Experimental results on a W-band UAV SAR system in strip-map mode indicate that this approach also outperforms other existing postprocessing autofocus methods.**

***Index Terms*— Motion estimation and compensation, symmetric triangle frequency-modulation continuous waveform (STFMCW), synthetic aperture radar (SAR) imaging.**

I. Introduction

In RECENT years, synthetic aperture radar (SAR) installed on unmanned aerial vehicles (UAVs) has been widely adopted in military and civilian applications due to its ability to achieve high resolution without time and weather restrictions [1], [2]. Generally, SAR collects echoes from the observed scene along a linear path with a constant forward velocity and coherently processes the signal to form an image [3]. For the SAR imaging formation process (IFP), there are four mainstream candidates: the range-Doppler algorithm, the chirp scaling algorithm, the backprojection algorithm, and the range migration algorithm (RMA) [4], [5]. Faced with the demands for real time and possible squint geometry in UAV SAR applications, the RMA can realize range cell migration correction (RCMC) and azimuth compression (AC) simultaneously without any approximation [6]–[8]. Therefore, it is regarded as an ideal imaging algorithm for UAV SAR systems.

In practice, UAVs usually fly at a relatively low altitude, and the SAR system inevitably suffers from air turbulence. Moreover, UAVs tend to be more sensitive to the effects of environmental vibration due to their lightweight and small size. As a result, the introduced motion errors, such as trajectory deviation, velocity, and altitude variation, will strongly impair the imaging quality, including geometric distortion and resolution losses [9]–[11]. Under high-resolution conditions, the negative effects can be more significant. On the one hand, the azimuth phase error (APE) will broaden the target impulse response function and increase the level of the sidelobe. On the other hand, nonsystematic migration through the range cell (MTRC) will disperse the energy of a target into several different bins. Particularly, during RMA processing, the Stolt interpolation performed in the wavenumber domain can destroy the linear relationship between the RCM and APE [7]. Furthermore, it can change the characteristics of motion error and enlarge its magnitude.

In fact, both airborne and spaceborne SAR face the problems of nonlinear trajectory imaging. Spaceborne SAR flies at a high altitude, so the forward acceleration in the azimuth direction can be regarded as constant in most cases, and variable PRF can address this issue properly [12]. There are a few differences in the modeling of slant range between spaceborne and airborne SAR. For the latter, the flight trajectory can be much more complex, and nonlinear characteristics can occur in three dimensions. To obtain a fine focused image result, the precise demand for navigation data is much stricter and higher, and algorithms based on novel interpolation, i.e., controlled singular value decomposition, can offer a solution [13]. However, for UAVs, limited by their payload, an inertial navigation system (INS) often has low accuracy and poor stability and cannot meet high-resolution imaging requirements [14], [15]. Therefore, motion estimation can be a crucial issue in UAV SAR imagery. Taking into account the sensitivity of the echo to

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the relative distance between the antenna phase center (APC) and the target, a strategy based on the returned signal can be an advisable solution [9].

To date, in the field of SAR imagery, there are many approaches to address the 2-D defocus based on returned signals. Commonly, two aspects, i.e., MTRC and APE, are considered separately and sequentially [16]–[19]. On the one hand, methods for residual RCM can deal with great motion errors but often with low precision. There are two primary alternative approaches. One is to apply the cross correlation (CC) function on the adjacent range profile and estimate the gradient of the RCM [20]. The estimation accuracy is at the subpixel level, and an upsampling operation is necessary to improve the precision. The other is to confine the RCM into one gate via downsampling and then obtain the APE at a coarser resolution, which can fail for heavy deviations with large variations [21], [22]. Different from these two algorithms, Fu *et al.* [23] proposed a method based on split-band signal interferometry, which has high accuracy for residual RCM without upsampling. Inspired by this idea, a novel algorithm based on symmetric triangle signals was presented in [24]. This method successively measures the range shift directly with each pair of up-ramp and down-ramp chirps to avoid the accumulation of errors. However, it should be noted that, in this method, although the RCM is sufficiently accurate, the original motion errors cannot be obtained because they have been modified after the Stolt interpolation. On the other hand, in terms of APE, the phase gradient autofocus (PGA) algorithm and its improvements are widely used, and its performance is dependent on the scene to a large extent [25]–[28]. Some 2-D autofocus approaches based on the RMA have been reported. Mao *et al.* [29] exploited the derived prior knowledge on the phase error structure. Lin *et al.* [30] proposed a space division algorithm to compensate for azimuth-dependent motion errors. In addition, in the polar format algorithm (PFA) framework, a cascaded correction based on subaperture division was presented in [31], and a novel algorithm based on spatial frequency fragmentation was developed in [32]. However, these methods can be applied only for a spotlight mode, rather than a strip-map mode. In summary, most existing autofocus algorithms use postprocessing; MTRC correction or APE compensation is separate and sequential. Especially for frequency-domain imaging algorithms, operations processed in the frequency domain, e.g., the Stolt interpolation, can change the form of motion errors in the echoes. Although the envelope of residual RCM or the curve of APE can be obtained by some existing methods, their physical implications are not very clear, and we can hardly recover the trajectory derivations. Moreover, isolated prominent point targets with a high signal-to-noise ratio (SNR) are commonly required, which brings challenges to obtaining fine-focused imaging quality and high-accuracy geographic information. Therefore, developing an effective data-driven motion estimation algorithm with a looser restriction on the scene remains a problem for UAV-borne SAR systems.

In this article, we develop a novel motion estimation method based on symmetric triangular linear frequency-modulation continuous microwave (STLFMCW) signals. As described in [33], NASA developed a Doppler LiDAR sensor with the STLFMCW signal for landing on Mars without the assistance of a global positioning system (GPS). Experiments suggest that this approach can provide reliable and high-accuracy velocity information for navigation, which has validated its potential to estimate motion parameters [34]. However, the side-look mode is complicated for conventional SAR imaging systems. On the one hand, from the perspective of velocity measurement, the velocity along the line-of-sight (LOS) is very small, and the difference between adjacent pulses is much less than one range resolution in most cases. Therefore, it is difficult to measure the velocity accurately through the CC method based on raw data. To improve the precision, upsampling in the range direction is necessary. Unfortunately, notwithstanding the heavy computational burden, the accuracy may not improve with an increase in the range oversampling rate [35]. For the sake of the whole residual RCM calculation, it is necessary to take an azimuth integral with respect to the range misalignment between adjacent range profiles. Consequently, estimation errors would propagate and further accumulate during this process. On the other hand, in terms of motion measurement, we need to track the envelope of the range compressed signal, meaning that the observed targets must be isolated in the range direction at least. Based on the above analysis, traditional methods for velocity and distance measurement in ground moving target indication (GMTI) maybe not suitable for motion estimation in SAR imaging [36], [37]. In our method, MTRC can be eliminated by signal interferometry between up-ramp and down-ramp chirp sections, leaving only the azimuth phase history. The reserved phase term consists of two parts. One is the inherent terms caused by normal trajectory, which mainly contains linear and quadric terms that are determined by the distribution of targets. The other part results from motion errors, and we can extract it from the signal after removing the inherent phase term. However, in practice, it is difficult to estimate the inherent items with high accuracy, and the residual errors can still defocus the final image. Fortunately, we have relatively abundant knowledge on the negative effect of these two low-frequency aberrations. Therefore, it is advisable to divide the phase term into three sections with respect to the order of errors in our approach, which runs a three-step estimation: 1) obtain the high-order phase errors through an integration of the detrended phase gradient; 2) solve the second-order term with the azimuth profile width minimization (APWM) algorithm in the azimuth wavenumber domain; and 3) remove the linear component by phase gradient matching and jointing.

Simulations verify that our proposed method performs well under conditions without ideal isolated point targets. Moreover, practice experiments acquired by a W-band strip-map UAV-borne SAR also demonstrate that our method has a high estimation accuracy with a large estimation range. Compared with the existing postprocessing strategy, the proposed method reduces the strict requirement for the scene to some extent. Distributed targets with acceptable SNRs that are relatively isolated in the azimuth dimension are needed, rather than ideal prominent points that must exist. As a consequence, we can extract reliable motion parameters from raw data and further

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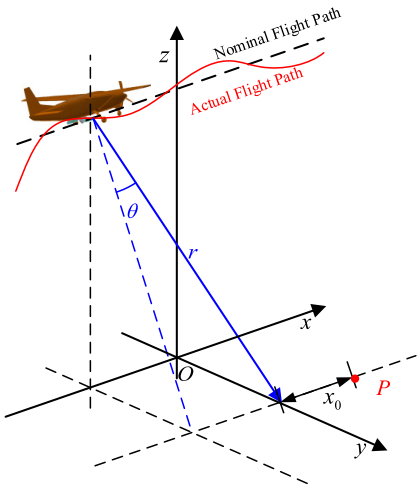


Fig. 1. Geometry of the SAR system.

achieve satisfactory imaging quality without losing positional accuracy. In addition, we should also note that this approach is applicable even without the assistance of external motion measurement equipment, such as the inertial measurement unit (IMU)/GPS. This is of great significance for system implementation in size minimization, weight saving, and cost reduction.

This article is organized as follows. In Section II, the signal model of STLFMCW SAR is first introduced, and the error model is carefully analyzed in the RMA process. Section III illustrates the principle of signal interferometry, and then, the implementation of the motion estimation algorithm is introduced in detail. The results of both simulations and practice experiments are provided and analyzed in Sections IV and V. Finally, the conclusions are drawn in Section VI.

II. Problem Statement

In this section, we first derive the motion error model during UAV SAR data collection, then establish the signal model of symmetric triangle FMCW SAR, and, finally, formulate the echo with motion errors in the RMA to facilitate the introduction of the proposed motion estimation algorithm.

A. Motion Error Estimation Model

Let us consider a generic target $P(r, x_0)$, where its distance to the ideal APC position is r and the ground distance from the scene center is x_0. Taking into account nonideal movements, the instantaneous range from the target to the actual APC can be expressed in (1) in the absence of any squint angle, i.e., $\theta = 0$. Fig. 1 shows the data collection geometry

$$\begin{aligned} R(X; r, x_0) &= \sqrt{(r + \Delta r(X; r))^2 + (X - x_0)^2} \\ &= \sqrt{r^2 + (X - x_0)^2 + 2r\Delta r(X; r) + \Delta r(X; r)^2} \\ &= R_0(X; r, x_0) + \Delta R(X; r, x_0) \end{aligned} \tag{1}$$

where $X = vt_a$, v is the forward velocity, t_a is the slow time, and $R_0(X; r, x_0) = (r^2 + (X - x_0)^2)^{1/2}$ is the ideal instantaneous range. $\Delta R(X; r, x_0)$ denotes the slant range displacement along the LOS, which is the vector sum of the along-track and across-track errors. The majority of motion errors can be attributed to the radial errors that are composed of dy and dz, which are the deviations of the APC in the Z- and Y-axes, respectively. Applying the Taylor approximation on $\Delta R(X; r, x_0)$, we can obtain the following:

$$\begin{aligned} \Delta R(X; r, x_0) &\approx \frac{\partial R(X; r, x_0)}{\partial \Delta r(X; r)} \Delta r(X; r) \\ &\approx \Delta r(X; r) \cdot \left(1 - \frac{(X - x_0)^2}{2r^2}\right) \end{aligned} \tag{2}$$

where $\Delta r(X; r)$, independent of the azimuth coordinate of targets, denotes the projection of $\Delta R(X; r, x_0)$ onto the yz plane. To avoid phase distortion, the following supposition is commonly used [10]:

$$\left| \frac{(X - x_0)^2}{2r^2} \cdot \Delta r \right| \le \frac{\lambda}{4\pi}. \tag{3}$$

That is, $\Delta r < (\lambda / 2\theta_{\text{bw}}^2)$, where θ_{bw} denotes the azimuth beamwidth.

In the SAR community, both strip-map and spotlight modes can possess comparable radar parameters and collect similar signal data. We commonly associate wide-area imaging at coarser resolution with a strip-map mode and reach a finer resolution for smaller scenes with the spotlight mode. In other words, compared with the strip-map mode, there would be a larger azimuth beamwidth in spotlight SAR with the same physical antenna. Considering the application of UAV SAR, the constraint condition in (3) can be easily satisfied for most cases in the strip-map mode. However, the assumption that the azimuth beam is narrow may be invalid in spot mode (especially for extremely high-frequency signals, e.g., the W-band), more easily losing the approximation that $\Delta R(X; r, x_0) \approx \Delta r(X; r)$. Therefore, the analysis and verification are primarily concerning the strip-map mode.

B. STLFMCW Signal Model

A symmetric LFMCW chirp consists of two sections with positive and negative slopes. The frequency centered at f_c spans the bandwidth B_r with chirp rate γ and then ramps back down. This up-to-down cycle is repeated at the PRF, giving a period of T_r. Regardless of the signal amplitude, the transmitted signal in the time domain can be expressed as

$$\begin{cases} s_t^+(t_r) = \text{rect}\left[\dfrac{t_r}{T_r}\right] \cdot \exp\left(j2\pi f_c t_r + j\pi\gamma t_r^2\right) \\ s_t^-(t_r) = \text{rect}\left[\dfrac{t_r - T_r}{T_r}\right] \cdot \exp\left(j2\pi f_c t_r - j\pi\gamma t_r^2\right). \end{cases} \tag{4}$$

According to the analysis in Section II-II-A, we extend the model with a squint angle θ, and the ideal instantaneous range can be

$$R_0(X; r, x_0) = \sqrt{(r\cos\theta)^2 + (X - r\sin\theta - x_0)^2}.$$

After baseband modulation, the positive section of the received signal is given by

$$\begin{aligned} s_r^+(t_r, X) = \sigma(r, x_0) \cdot p_r(t_r) \cdot p_a(X) \\ \cdot \exp\left[j\pi\left(2f_c(t_r - \tau) + \gamma(t_r - \tau)^2\right)\right] \end{aligned} \tag{5}$$

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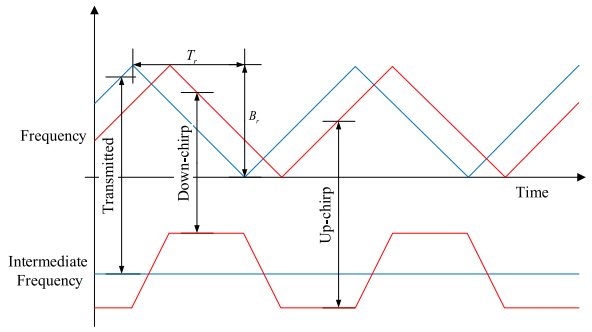


Fig. 2. Transmitted and corresponding received LFMCW signal and the resulting intermediate frequency of the homodyne signals.

where $\tau = \tau_0 + \Delta\tau = 2R/c$ is commonly called round-trip time. Note that $\sigma(r, x_0)$ is the complex weight corresponding to the scatter reflectivity, $p_r(t_r)$ is the window in the range direction, and $p_a(X)$ denotes the azimuth amplitude modulation. Herein, these three items would be neglected for briefness since they are inessential for our scope of the study.

In the dechirped receiving mode, the transmitted signal can serve as a reference to mix with returned data and generate an interference signal. The time–frequency relationship of the STLFMCW signal is depicted in Fig. 2.

After removing the well-known residual video phase, the full expression of the signal in the wavenumber can be written as [6]

$$\begin{cases} s_{\text{if}}^{+}(K_r, X) = \exp\left(-jK_r^{+}R_1\right) \\ s_{\text{if}}^{-}(K_r, X) = \exp\left(-jK_r^{-}R_2\right) \end{cases} \tag{6}$$

where

$$\begin{cases} K_r^{+} = K_{rc} + \Delta K_r = 4\pi f_c/c + 4\pi f_r/c \\ K_r^{-} = K_{rc} - \Delta K_r = 4\pi f_c/c - 4\pi f_r/c \end{cases}$$

with a substitution $\gamma t_r \to f_r$. R_1 and R_2 represent the slant range from the same target for the up-ramp chirp and down-ramp chirp, respectively.

The triangular-modulated wave signal is a type of timing signal, and there is only one channel. The up-ramp chirp and the down-ramp chirp are sent to the ground in a temporally alternating sequence. The time offset between the two adjacent signals is equal to the pulse repetition interval (PRI), which is commonly on the level of 10^{-4} s. It is worth noting that the two types of signals are processed individually, and the difference induced by PRI on trajectory deviations is too small to be considered in practice. Therefore, it is reasonable to consider that they have experienced the same motion errors during flight. Based on the strong symmetry between the two sections in the STLFMCW signal, we formulate a unilateral signal with the same slant range and range wavenumber notations, i.e., R and K_r, in the following analysis:

$$s(K_r, X) = \exp[-jK_r R(X; r, x_0)]. \tag{7}$$

C. Motion Error Analysis in RMA

Conventional SAR IFP assumes that the platform moves at a constant forward velocity in a straight line. The APC of the SAR system mounted on a minisized UAV can easily deviate from the nominal flight path because of air turbulence and unbalanced motor power delivery. The presence of trajectory displacement introduces errors to the collected data and provides distortions to defocused SAR images. In this section, we analyze the effects of motion errors on SAR signals in RMA. Note that the standard RMA for a dechirped FMCW signal mainly consists of several essential steps, i.e., azimuth Fourier transform (FT), reference matched filtering, the Stolt interpolation, azimuth inverse FT (IFT), and range FT [6]. Herein, a modified algorithm is executed for the sake of being readily understood.

1) Azimuth FT: First, based on the principle of stationary phase (POSP), azimuth FT is performed to obtain the 2-D point target reference spectrum (PTRS), whose phase can be expressed as

$$\Phi(K_r, K_x) = -r\cos\theta\sqrt{K_r^2 - K_x^2} + K_x(x_0 + r\sin\theta) - K_r\Delta R\left(X^*\right) \tag{8}$$

where

$$X^* = -r\cos\theta\frac{K_x}{\sqrt{K_r^2 - K_x^2}} + x_0 + r\sin\theta$$

and the last term arises from motion errors.

2) Modified Stolt Interpolation: To investigate the impacts of Stolt on the signal in the range dimension, we apply a modified mapping on (8) to realize RCMC only, which is defined as [7]

$$\cos\theta\sqrt{K_r^2 - K_x^2} \to \Delta K_y + \cos\theta\sqrt{K_{rc}^2 - K_x^2}. \tag{9}$$

Then, the phase spectrum of signal is

$$\Phi_{\text{MST}}(K_r, K_x) = -\phi_e(K_r, K_x) - K_x(x + r\sin\theta) - r\left(\cos\theta\sqrt{K_{rc}^2 - K_x^2} + \Delta K_y\right) \tag{10}$$

where the phase error is

$$\phi_e(K_r, K_x) = \sqrt{\left(\Delta K_y\sec\theta + \sqrt{K_{rc}^2 - K_x^2}\right)^2 + K_x^2\Delta R\left(X^*\right)}.$$

3) Azimuth IFT: After performing an azimuth IFT by neglecting the effect of the motion errors on the SP point, the phase spectrum of the signal can be given by

$$\Phi_{AT}\left(\Delta K_y, X\right) = -K_{rc}\sqrt{(r\cos\theta)^2 + (X - x_0 - r\sin\theta)^2} - r\Delta K_y - \phi_e\left(\Delta K_y, K_x^*(X)\right) \tag{11}$$

where the approximated SP point is

$$K_x^* = -K_{rc}\frac{X - r\sin\theta - x}{\sqrt{(r\cos\theta)^2 + (X - r\sin\theta - x)^2}}.$$

The last exponential term resulting from errors can be expanded into the first-order Taylor series around $\Delta K_y = 0$, i.e.,

$$\phi_e\left(K_x^*, \Delta K_y\right) = \zeta_0(X) + \Delta K_y\zeta_1(X)\ldots \tag{12}$$

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in which we let θ' be the instantaneous squint angle

$$\zeta_0(X) = -K_{rc}\Delta R(X) \tag{12.a}$$

$$\zeta_1(X) = -\frac{\cos\theta'}{\cos\theta}\Delta R(X) - \frac{\tan\theta'}{\cos\theta'}\frac{\partial \Delta R(X)}{\partial X}r. \tag{12.b}$$

Details can be found in Appendix A. At this point, some comments on the above derivations are summarized.

1) In (12), $\zeta_0(X)$ denotes the APE and $\zeta_1(X)$ the residual RCM. After range compression with an FT along the range direction, the envelope error is determined by $\zeta_1(X)$. Note that the contribution of second- and other high-order terms with respect to ΔK_y, which are related to the range defocus, is ignored.
2) *For (12.a):* In the above processing, there are two main operations in the azimuth dimension, i.e., an FT and an IFT. Since the modified Stolt does not change the azimuth phase spectrum, the original motion errors in the time domain are reserved in the phase history, which can be utilized for motion estimation.
3) *For (12.b):*
   a) In the above analysis, the approximated SP point is obtained by ignoring the effects of motion errors when applying POSP. That is, the above expression is for approximate calculation only.
   b) The expression of residual RCM is much different from (12.a), meaning that the modified Stolt mapping in the 2-D wavenumber domain has changed the characteristics of motion parameters. The residual RCM is not only determined by the motion error but also related to its derivative, i.e., the across-track velocity.
   c) There is a range-dependent coefficient, i.e., r in the second term. Since the slant range is typically much larger than motion errors, it can magnify the negative influence of the inaccurate SP point.
   d) In the case of great motion displacements with heavy variations, the results obtained by POSP may be unreliable, and the deduced expression may be inexact. Therefore, the flight trajectory may not be recovered from the acquired RCM.

Therefore, for RMA, existing postprocessing algorithms cannot extract the original motion errors induced by flight derivation from the envelope of the signal. In addition, under conditions of the wide swath, the range-dependent effects on residual RCM cannot be neglected. Bulk compensation for residual RCM is insufficient, and some methods aimed at range-independent errors can only improve the image quality to a certain extent.

III. Algorithm Implementation

A. STLFMCW Signal Interferometry

The STLEMCW signal for SAR in the range-wavenumber and azimuth-time domains can be expressed as

$$\begin{cases} s^+(K_y, X) = \exp\left[-j\left(\Delta K_y R_{\mathrm{MS}} + K_{rc}\phi_{\mathrm{AP}}\right)\right] \\ s^-(K_y, X) = \exp\left[j\left(\Delta K_y R_{\mathrm{MS}} - K_{rc}\phi_{\mathrm{AP}}\right)\right] \end{cases} \tag{13}$$

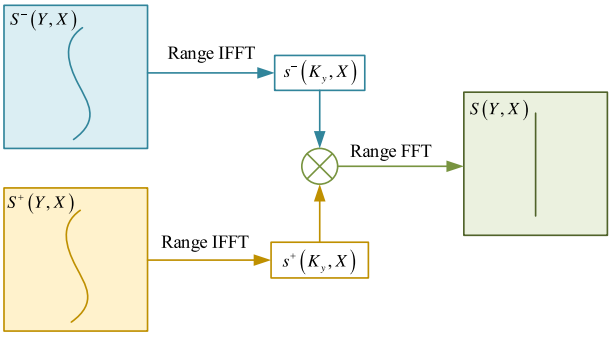


Fig. 3. Signal interferometry scheme.

where R_{MS} represents the envelope of the signal

$$R_{\mathrm{MS}} = r + \zeta_1(X) \tag{13.a}$$

and ϕ_{AP} denotes the phase history

$$\phi_{\mathrm{AP}} = \sqrt{(r\cos\theta)^2 + (X - x - r\sin\theta)^2} + \Delta R(X). \tag{13.b}$$

From (13), we can learn that the two sections of the signal have the same characteristics of errors but differ from the sign of ΔK_y. The phase information in the range direction can cancel each other out via signal interferometry, i.e.,

$$s(K_y, X) = s^+(K_y, X)\cdot s^-(K_y, X) = \exp(-j2K_{rc}\phi_{\mathrm{AP}}) \tag{14}$$

and the original motion errors in APE can be extracted, i.e.,

$$\Delta R(X; r, x_0) = \frac{\arg\left[s(K_y, X)\right]}{-2K_{rc}} - \sqrt{(r\cos\theta)^2 + (X - x_0 - r\sin\theta)^2}. \tag{15}$$

Note that the envelope of the interferometric signal can be a straight line after a range FT because there is no information about the range wavenumber. The graphic illustration can be provided in Fig. 3.

For all targets in the observed field, a common and practicable way to eliminate this term is based on an azimuth matched filter in the frequency domain on account of the similar structural characteristics of the returned signal in the range-compressed azimuth-frequency domain. The consistency rule of motion errors in the time domain would be broken after an azimuth FT, much less reconstruction of the original motion parameters. In the time domain, based on the abovementioned principle, for prominent point targets whose coordinates are known beforehand, the second additional term in (15) can be compensated easily and properly. In the case of only a distributed target exists, the critical challenge is to eliminate the inherent terms determined by its distribution, which consists of quadric and linear components. With the knowledge of the structure of the extra term, information on distributed targets with acceptable SNR can also be exploited.

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B. Motion Error Estimation

Based on the interferometry signal in (14), we present a novel approach to motion error estimation for the strip-map mode, which includes three steps.

Suppose that there are several high-quality targets that can be served for estimation. Each target can provide an observation aperture. There is a sequential loop in the first two steps to estimate the second- and higher-order components. The sequential loop will not be finished until all observation apertures are complete. The last step is to restore the remaining linear portion in each aperture. The technical contents of these parts will be explained in detail in the following.

1) High-Order Error Estimation: To obtain the high-order motion errors, we first rewrite the phase history in (13), i.e.,

$$\phi_{\mathrm{AP}}(X) = aX^2 + bX + c + Q_h(X) \tag{16}$$

where a, b, and c represent the coefficients of the low-order components and $Q_h(X)$ represents the third-order and upper order components. The derivative of $\phi_{\mathrm{AP}}(X)$ with respect to X can be expressed as

$$\frac{\partial \phi_{\mathrm{AP}}(X)}{\partial X} = 2aX + b + \frac{\partial Q_h(X)}{\partial X}. \tag{17}$$

We can remove the mean value or linear trend from (17) and obtain the gradient of the high-order motion errors. Then, the high-order component is achieved by integration, i.e.,

$$Q_h(X) = \int \mathrm{detrend}\left(\frac{\partial \phi_{\mathrm{AP}}}{\partial X}\right) dX. \tag{18}$$

It should be noted that, in (18), the normal linear and second-order components are both removed. Next, we need to recover these two portions.

2) Quadric Error Estimation: To estimate the quadric component, we proposed an APWM algorithm based on differential signals in (15). Assuming that there is no motion error left, the signal in (14) can be written as

$$s(K_y, X) = \exp\left(-j2K_{rc}\sqrt{(r\cos\theta)^2 + (X - x_0 - r\sin\theta)^2}\right). \tag{19}$$

By applying azimuth FT with POSP (see Appendix B), we can obtain the phase of reference spectrum, i.e.,

$$\Phi(K_y, K_x) = -r\cos\theta\sqrt{(2K_{rc})^2 - K_x^2} - K_x(r\sin\theta + x_0). \tag{20}$$

Neglecting the last linear term, we can generate an azimuth filter with the conjugate of (22), i.e.,

$$H(K_y, K_x) = r\cos\theta\sqrt{(2K_{rc})^2 - K_x^2}. \tag{21}$$

Then, followed by multiplication with the matched filter in (23), the signal can be compressed into a sinc-shaped impulse function after an azimuth FT. On this basis, we introduce the concept of azimuth profile (AP) into motion estimation and take the AP 3-dB width as the metric of the iteration process. If the residual phase errors are estimated accurately, the width will converge to a minimum.

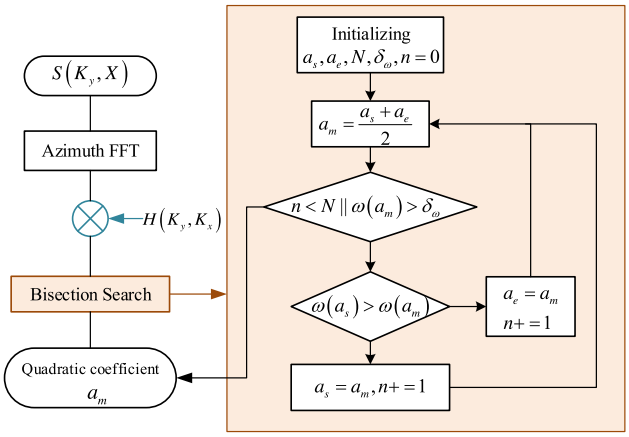


Fig. 4. Flowchart of APWM.

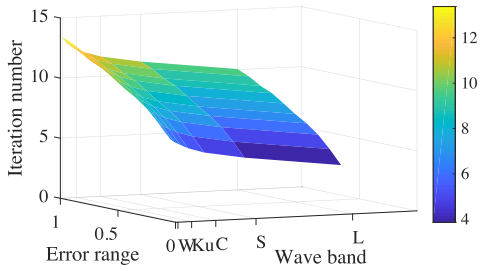


Fig. 5. Number of iterations.

The quadric motion error in (16) can be expressed as

$$Q_2(X) = aX^2. \tag{22}$$

We can achieve APWM by a 1-D search for the second-order coefficient, i.e., a.

Herein, with the basic concept of the bisection method, we design a modified search strategy [38]. Fig. 4 shows the flowchart, in which a_s and a_e are the start and end of the search, a_m is the median, N is the maximum number of iterations, ω represents the width of the AP, and δ_ω is the threshold of the APW. The algorithm will stop within finite iterations when the APW converges to its minimum. To investigate the number of iterations, we provide the following analysis. Generally, it is common practice to use $\pi/4$ rad as the acceptable threshold for phase error. Suppose that the maximum absolute value of motion error is ΔR_m, and set the hunting zone to $[-(\Delta R_m/2), (\Delta R_m/2)]$; we can obtian

$$N \le \log_2\left(\frac{4\Delta R_m K_{rc}}{\pi}\right) + 1. \tag{23}$$

From (23), we can learn that the number of iterations is largely determined by the wavelength and the scope of trajectory deviations. The quantitative analysis is provided in Fig. 5, from which we can see that the smaller the wavelength and the larger the motion errors are, the more iterations there are. For the W-band signal, the number of iterations can be limited to 14 when the maximum value of errors is within 1 m.

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TABLE I
SYSTEM PARAMETERS

Parameter	Value
Carrier frequency	94GHz
Band width	2GHz
Time width	100us
Flight height	100m
Forward velocity	8m/s
Range cell	7.5cm
Azimuth cell	1.6mm

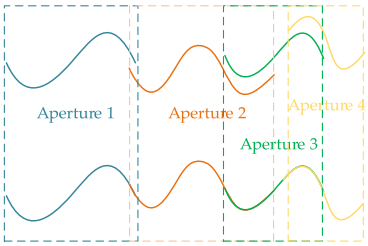


Fig. 6. Graphic illustration of the error gradient matching for adjacent apertures.

3) Linear Error Estimation: Because the linear phase error does not impair the azimuth focus for each target, there would be a residual linear component for each estimation aperture after implementing the aforementioned two steps. In the case of a large linear term, MTRC may occur although it does not attribute to the spread of APW. Because the linear error corresponds to a constant cross-track velocity, we can eliminate it by gradient curve matching for adjacent apertures and then let the mean value of the error gradient during the whole observation time equal zero. Fig. 6 provides the graphic illustration.

This approach can always work well because the normal fitting flight path follows a linear rule along the azimuth direction, and it is expected that there is no across-track motion. In addition, we should also note that the details of high-order motion errors acquired by low-accuracy IMU may be lost, while the low-order components can also be a candidate for the residual linear estimation in each observed aperture.

IV. SIMULATIONS

To validate the developed signal model and the performance of the proposed algorithm, some results based on simulated data are provided in this section. In the point target simulation, the built differential signal model is verified. In the distributed target simulation, the letter "E" is set to verify the performance of the whole estimation process. The SAR system parameters are listed in Table I.

A. Point Target Simulation

In this part, we simulated the echo signal from a prominent point scatter. There is a 2-D defocus in the SAR image caused by considerable motion errors.

To prove the effects of the Stolt interpolation on motion errors in the signal, the residual range migration errors are obtained by tracking the peak position of the signal in the range-compressed and azimuth-time domain, as depicted in Fig. 7(a). Recalling the proposed method, Fig. 7(b) shows that the envelope of the interferometric signal generated by (18) is a straight line, from which we can learn that the range information is completely canceled out and the reserved APE can be easily retrieved by tracking the phase history. Therefore, with the knowledge of the target geographical coordinates, the second normal phase function in (15) can be compensated easily. One thing that is worth mentioning is that phase wrapping may occur due to the short wavelength. Generally speaking, it is not a difficult thing to unwrap and recover its true value from $[-\pi, \pi]$. Herein, we suppose the phase difference between two adjacent cells is no larger than 2π, which would not introduce errors in phase extraction. That is, $2K_{rc}(\partial \Delta R/\partial t_a) \cdot 2T_r \leq 2\pi$; then, $(\partial \Delta R/\partial t_a) \leq (\lambda/8T_r)$. In practice, experimental evidence suggests that this assumption is available in most cases, while, for extremely bad environmental conditions, we can still guarantee it by coarse focusing based on low-accuracy motion measurement sensors. Fig. 7(c) gives a comparison of motion errors in the range direction; the solid green line represents the residual RCM in Fig. 7(a), the red dashed line represents the added motion errors, and the estimation results by extracting APE from the interferometric signal in Fig. 7(b) are colored blue. These results demonstrate that RCMC with the Stolt interpolation in the range wavenumber domain will change the characteristics of the motion error; in addition, the envelope and phase no longer remain synchronized. As depicted in Fig. 7(d), the maximum absolute estimation error is within 1×10^{-4} m, which is far less than $\lambda/16$, guaranteeing the precision for image focusing [39].

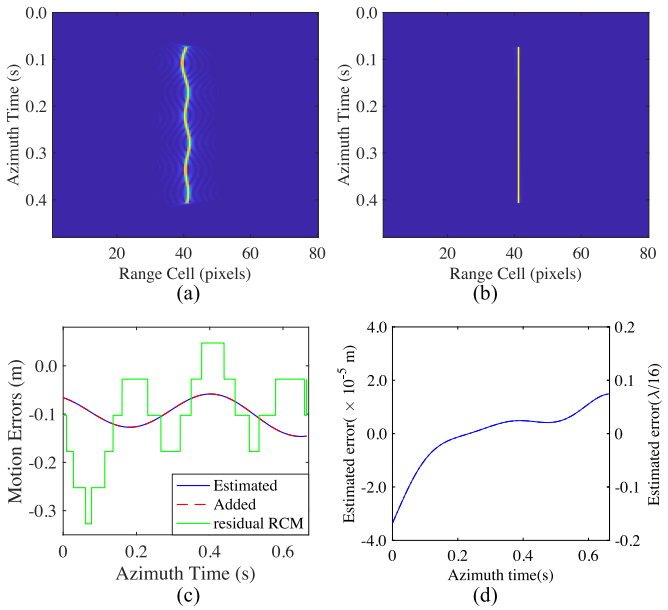


Fig. 7. (a) Envelope of the range-compressed signal. (b) Envelope of the interferometric signal. (c) Comparison of the residual RCM, estimated motion errors, and ideal motion errors. (d) Difference between the estimated and ideal motion errors.

Taking Gaussian white noise into consideration, the SNR can be defined as

$$\mathrm{SNR} = \frac{A^2 N}{\sigma_n^2} \tag{24}$$

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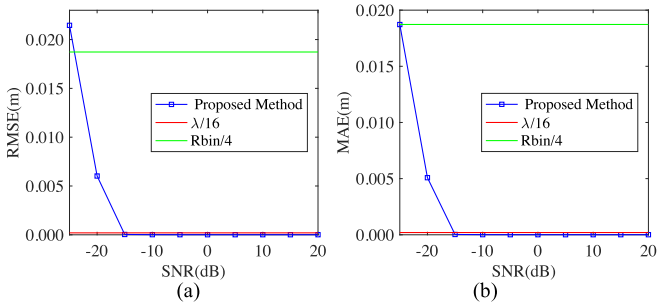


Fig. 8. Estimation performance analysis versus SNR. (a) MAE. (b) RMSE.

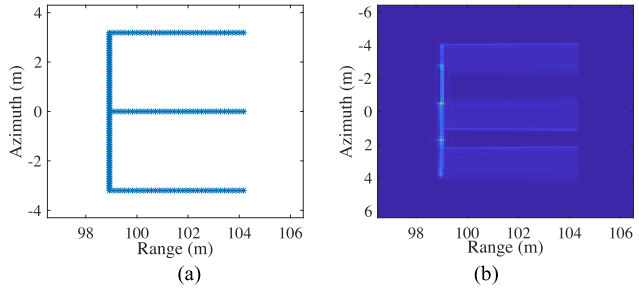


Fig. 9. (a) Scatter model of a distributed target. (b) Image result with motion errors.

where A is the signal amplitude, N is the length of the signal, and σ_n^2 is the variance of the noise. With the purpose of presenting robustness, we perform 100 Monte Carlo simulations at each SNR level. In Fig. 8, the horizontal axis is the SNR of the received signal, and the vertical axis is the mean absolute error (MAE) and the root mean squared error (RMSE) of the estimation result of motion errors. The proposed method shows excellent performance under high SNR conditions. The degradation of SNR will naturally cause a reduction in estimation accuracy. However, when the SNR reaches -25 dB or higher, the RSME is less than a quarter of one range cell, validating the effectiveness and robustness of our proposed method.

B. Distributed Target Simulation

With the system parameters in Table I and the target feature in Fig. 9(a), symmetric triangle FMCW SAR data injected with motion errors are generated in the dechirp receiving mode. The image result without motion compensation (MoCo) is shown in Fig. 9(b). The image is seriously blurred, and its structural characteristics are lost entirely. We cannot isolate a prominent point from the target, meaning that existing methods based on points are no longer valid. However, with respect to the energy distribution, we can segment the image into three apertures along the azimuth direction, and then, our developed method can be applied to realize motion estimation.

Some important intermediate results in the full flow are given as follows. The estimated high-order motion error is shown in Fig. 10. Due to the existence of low-order components, these error curves do not coincide in the overlapped slow time. After removing high-order motion errors, there are only second- and first-order elements attributed to the

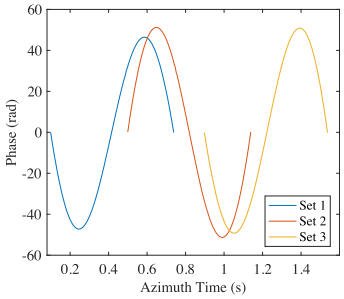


Fig. 10. High-order motion errors.

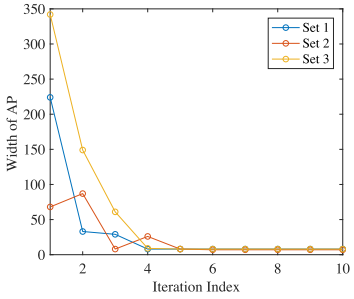


Fig. 11. Quadric component estimation with the APWM method.

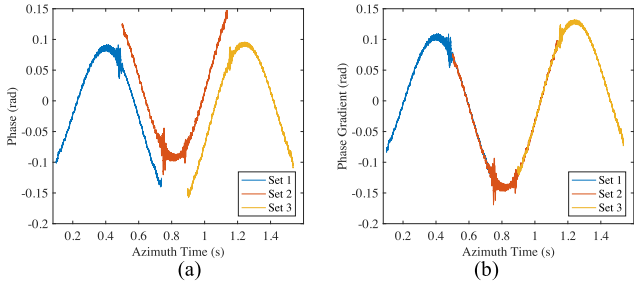


Fig. 12. (a) Gradient of motion errors with linear errors. (b) Gradient of motion errors without linear errors.

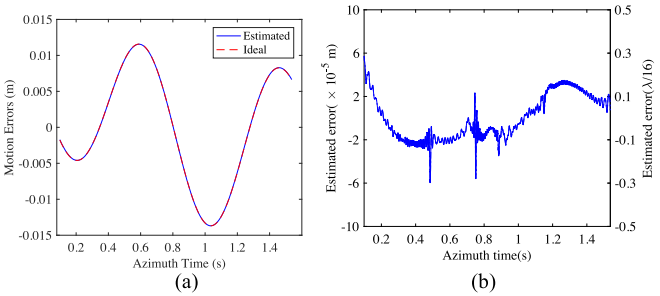


Fig. 13. (a) Comparison of motion errors. (b) Estimation errors.

normal path and the flight derivation. To estimate the quadric component, a novel APWM method is performed. By means of bisection, the 3-dB APW versus iteration times of the proposed algorithm is depicted in Fig. 11. Numerical results show rapid convergence of the iterative procedure.

Fig. 12(a) presents the estimated gradient, in which these three curves have the same characteristics but varying value ranges. Finally, with the last estimation for the linear

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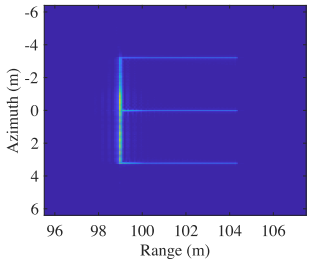


Fig. 14. Imaging result integrated MoCo with our proposed method.

TABLE II
SYSTEM PARAMETERS

Parameter	Value
carrier frequency	94 GHz
band width	2 GHz
Time width	200 us
flight height	100 m
forward velocity	8 m/s
Range cell	7. 5×10^{-2}m
Azimuth cell	3.2×10^{-3} m

component with gradient matching, the difference is removed in Fig. 12(b). In Fig. 13(a), the blue solid line and the red dashed line represent the ideal motion error and the estimated motion error, respectively. The difference between the two (normalized by $\lambda/16$) is colored with blue, as depicted in Fig. 12(b). The imaging result compensated with estimated motion errors is presented in Fig. 14, from which we can see more details of the distributed target. These results suggest that the proposed algorithm is available even if no prominent point exists.

V. EXPERIMENTS

The continuous turning of the radar beam to direct at the target area during the flight in spotlight mode adds to the hardware complexity, and it is much more difficult to implement, especially for the small UAVs. In this section, practice experiments in the strip-map mode are designed to verify the effectiveness of the proposed approach. The raw data are acquired by a W-band UAV SAR STLFMW SAR system mounted on a UAV, as shown in Fig. 15 [40]. It is worth mentioning that this prototype of the radar system has a weight of 400 g and a volume of 67 mm × 60 mm × 50 mm. The antenna beamwidth is 12° × 3° (range × azimuth). Some important system parameters are listed in Table II.

The radar platform is equipped with an IMU that can provide motion information at a velocity accuracy of 0.02 m/s and a position accuracy of 0.05–0.3 m (after postprocessing) at 200 Hz and a GPS system whose output rate is 5 Hz. The relative accuracy and the updating rate can hardly satisfy the imaging demand. Therefore, after the traditional compensation based on the measured auxiliary data, the quality of the image still suffers from the residual motion errors, and the autofocus operation based on raw data is indispensable to form high-quality images. In the following experiments, we note that, in the cases of both medium and severe motion errors,

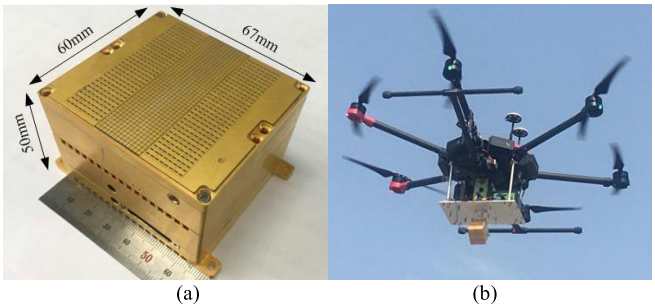


Fig. 15. (a) Mini-SAR system prototype. (b) Multirotor UAV platform.

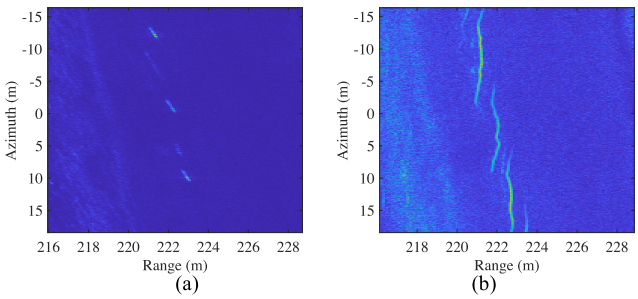


Fig. 16. Image results with original motion errors in (a) range-compressed azimuth-time domain and (b) full-compressed domain.

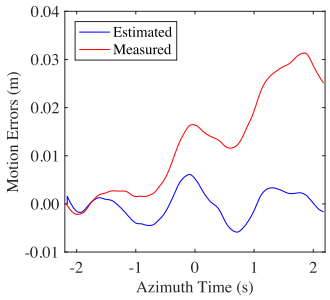


Fig. 17. Comparison of motion errors between the measured data with IMU/GPS and our estimation results.

the proposed method can precisely estimate them with high accuracy, which is enough to remove 2-D defocus, including APE and nonsystematic RCM. Finally, well-focused images can be generated by using the traditional compensation method on returned echoes with our estimated motion parameters.

A. Proposed Motion Estimation Performance

To demonstrate the capability of the proposed estimation algorithm, an experiment is performed on raw data containing motion errors. Without any auxiliary motion measurement data, the original image merely based on the RMA imaging process can be defocused with a large degree of blurring in both two dimensions. For the sake of comparison, we provide the motion errors acquired by our proposed method and the equipped measurement sensors.

As depicted in Fig. 16(a), the envelope of the signal in the range-compressed domain after modified Stolt interpolation exceeds more than one range cell. Consequently, the fully compressed image is subjected to severe resolution loss not

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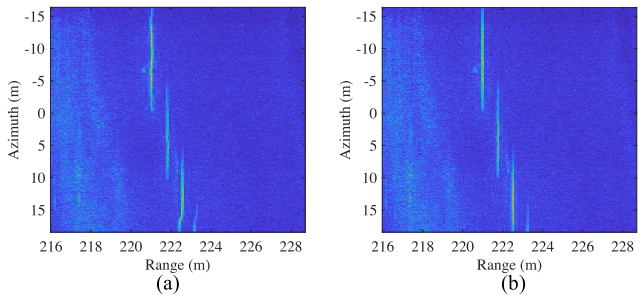


Fig. 18. Images compensated with motion errors acquired by (a) measured data based on IMU/GPS and (b) estimated results based on our proposed method.

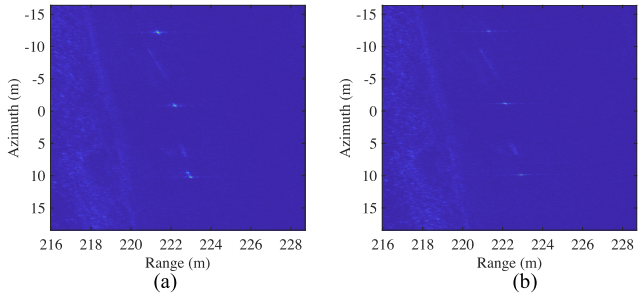


Fig. 19. Images compensated with motion errors acquired by (a) measured data based on IMU/GPS and (b) estimated results based on our proposed method.

only in the azimuth but also in the range dimension in Fig. 16(b). That is, the dominant blur is in both two directions.

Fig. 17 shows the comparison result of motion errors, in which the red line represents the data measured by equipped sensors and the blue line denotes the estimation result. There is a relatively great difference not only in the value range but also in the variation trend. To distinguish the more accurate results, we perform the traditional compensation strategy with the two groups of motion errors. Fig. 18 gives the comparison of range-compressed image results, in which Fig. 18(a) represents the results with measured data and Fig. 18(b) is with our estimated motion parameters. There is still an obvious MTRC in Fig. 18(a). The signal envelopes of targets in Fig. 18(b) are all corrected properly into "straight lines," meaning that our proposed method can achieve higher accuracy.

Correspondingly, both fully compressed images are shown in Fig. 19(a) and (b). For the sake of a close look, Fig. 20 presents the zoomed point targets with upsampling along the range direction, in which Fig. 20(a)–(c) shows the imaging results compensated with motion measurement sensors. There are significant APEs, resulting in energy dispersion and resolution loss. In severe cases, one target would be split into two. In contrast, the right column depicts the results obtained by our proposed method. It is obvious that we can achieve a higher estimation accuracy of motion errors and a better focusing quality subsequently. Furthermore, we introduce three terms, i.e., the 3-dB impulse response width (IRW), the peak sidelobe ratio (PSLR), and the integral sidelobe ratio (ISLR), to describe the focusing quality of point targets in the scene, as listed in Table III. Compared with the imaging results based on measurement sensors, it seems that few arguments

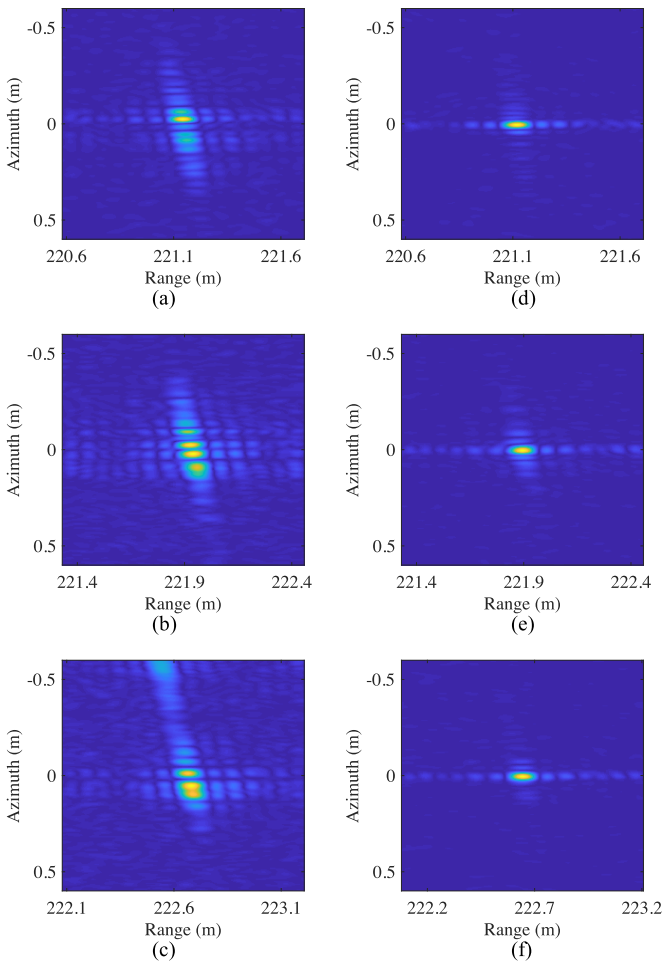


Fig. 20. Comparison of the focusing quality: (a)–(c) are compensated with motion parameters by the IMU/GPS system and (d)-(f) are with motion errors obtained by our method.

(such as IRW in the range direction) based on the proposed method are similar or even show degradation. We should note that severe azimuthal energy spreading, especially in Fig. 20(a), can result in two main lobes. To conclude, these evaluation results validate the effectiveness of the motion estimation method.

B. Comparison of MoCo Results

In this section, we perform experiments on another real measured dataset acquired by the abovementioned SAR system. Moreover, comparisons with other postprocessing autofocus algorithms are also presented.

In the process of the RMA, the APC is assumed to move following a straight flight trajectory with a constant forward velocity. Therefore, the produced imagery after traditional IFP suffers from serious 2-D defocus, which can be clearly seen in the original image result, as shown in Fig. 21(a). In this case, the standard PGA faces difficulties due to the existence of MTRC. In [20], a postprocessing method was developed, which consists of two main steps: the first step corrects the RCM based on the CC method, and the second step compensates for the residual APE. Fig. 21(b) gives the fully compressed results with 15-fold upsampling in the range direction. The image quality is improved, but blurring still

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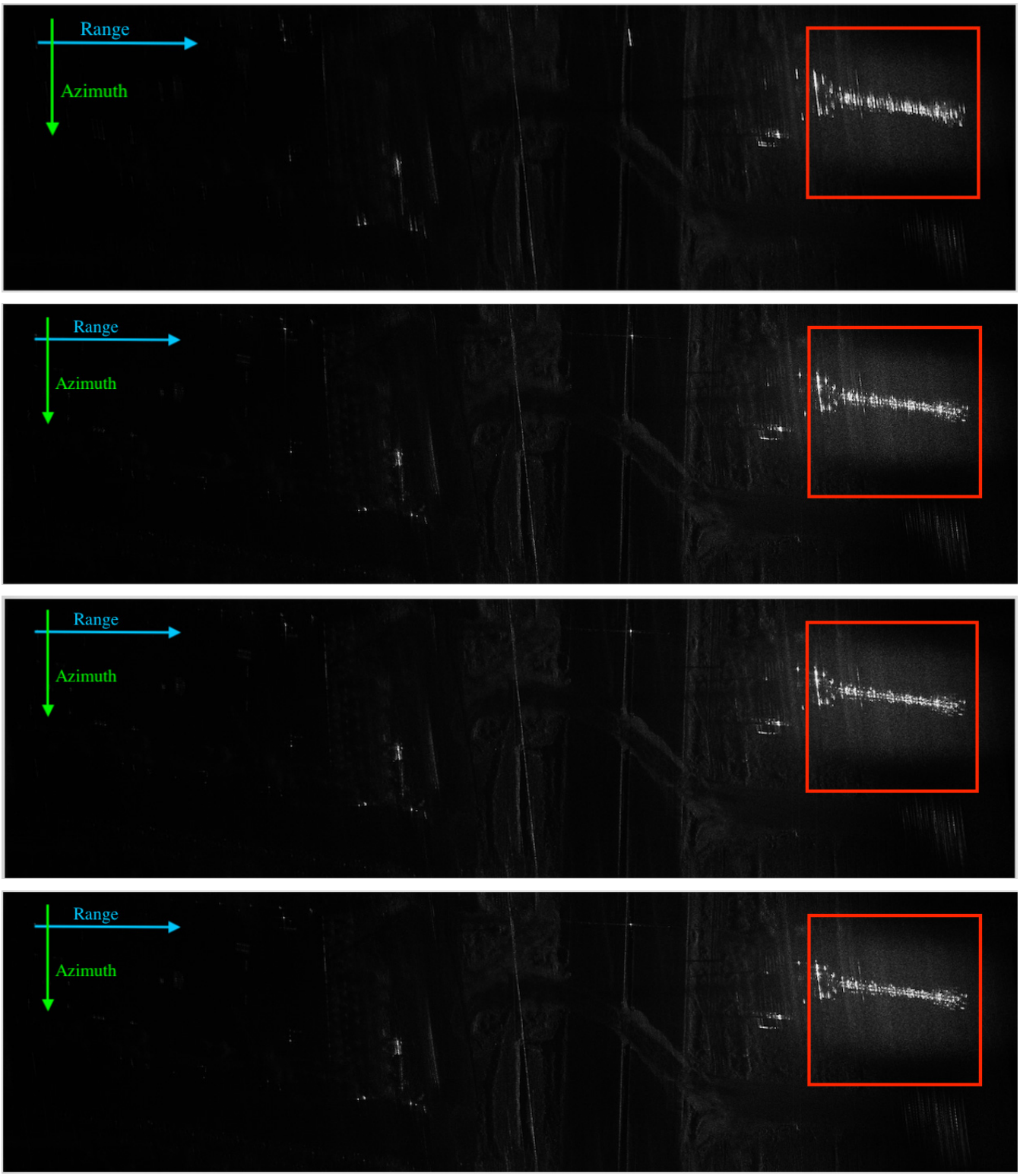


Fig. 21. Comparison of imaging results: (a) with motion errors and compensated with (b) postprocessing in [20], (c) postprocessing in [24], and (d) proposed method.

occurs, for which there are three dominant reasons. One is that the CC-based method needs an upsampling operation to reach a subpixel level, and the accuracy may not always be improved with an increasing oversampling rate. Moreover, the gradients between adjacent pulses are too small to measure precisely, and the accumulative errors inevitably propagate, especially for complex and constantly changing circumstances after a long aperture time. The last explanation is that the large scale of errors can impair the SP point, and the range-dependent errors after the Stolt interpolation will reduce its

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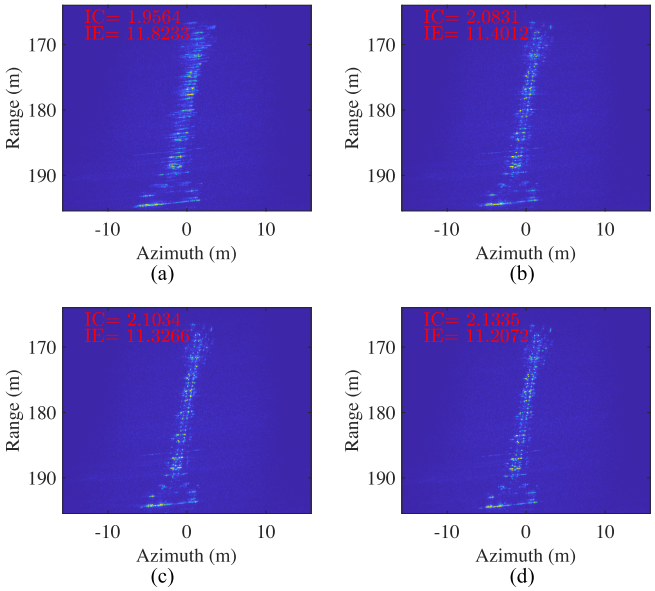


Fig. 22. Comparison of local-imaging results in Fig. 21: (a) with motion errors and compensated with (b) postprocessing in [20], (c) postprocessing in [24], and (d) proposed method.

effectiveness to some extent. As a comparison, a novel post-processing MoCo algorithm based on STLFMCW is presented in [24]. In this method, up-chirp and down-chirp signals are both split into two subbands. Then, one STLFMCW signal consists of four portions, and MTRC can be acquired by signal interferometry, which involves range envelope tracking. Compared with the former, the method in [24] has a higher accuracy with a lower computational burden. Fig. 21(c) depicts the corresponding result, in which a more satisfactory focusing quality is obtained. There are two main issues worth noting. On the one hand, the estimated RCM has been modified by Stolt interpolation, from which we can hardly solve the motion errors caused by flight deviation. On the other hand, the linear relationship between RCM and APE has been destroyed, making the compensation for range and azimuth independent. That is, although MTRC can be properly removed, residual APE remains, and the PGA is commonly used. Because the PGA principally relies on prominent isolated points, the focusing performance is limited by the observed scene to a large extent. Fortunately, our proposed method can still achieve a fine focused image, as shown in Fig. 21(d). Although our eyes cannot tell a significant difference between Fig. 21(c) and (d), we should note that it has additional benefits. For example, we can retrieve the motion parameters and compensate for the residual RCM and APE simultaneously before the Stolt interpolation. To closely examine the details, the corresponding zoomed-in version of the rectangular area in Fig. 21 containing a signal tower is magnified in Fig. 22. These imaging results show that 2-D defocus can be properly eliminated by our proposed algorithm. There is hardly a standard control point. To further compare and quantify evaluate the focus quality, the image contrast (IC) and image entropy (IE) are also calculated and listed in Table IV, from which we learn that the performance of the proposed algorithm achieves more significant improvements than other existing postprocessing methods in refocusing.

TABLE III
FOCUSING PERFORMANCE COMPARISON

Target point	Motion source	PSLR (dB)		ISLR (dB)		IRW(m)	IRW(m)
		Range	Azimuth	Range	Azimuth	Range	Azimuth
1	measured	-13.22	-4.54	-10.91	1.00	0.0796	0.0223
	estimated	-13.45	-14.10	-10.96	-11.81	0.0796	0.0235
2	measured	-12.57	-0.33	-9.83	3.42	0.0796	0.0743
	estimated	-12.78	-11.78	-10.18	-9.36	0.0796	0.0271
3	measured	-12.68	-1.32	-9.66	1.57	0.0796	0.1315
	estimated	-13.02	-16.81	-10.43	-14.83	0.0796	0.0255

TABLE IV
FOCUSING PERFORMANCE COMPARISON

Image	IE	IC
Without compensation	13.7834	2.2803
With post-processing in [20]	13.3789	2.3886
With post-processing in [24]	13.3101	2.4050
With the proposed method	13.2112	2.4137

There is one thing that should be noted is that, due to the low altitude of UAV, the range swath of the SAR system is relatively small, and the range-dependent errors can be ignored in practice. In the abovementioned experiments, the majority of slant range errors in the scene can be removed properly with bulk compensation. Nevertheless, we cannot deny that the assumption of range-invariance has constraints. In the case of wide swath or heavy errors, imaging results would suffer from space-variant errors. Under this condition, more targets are required, and limited range block division can be an alternative approach to solve it.

VI. CONCLUSION

Faced with difficulties in UAV SAR imagery, the RMA can be regarded as a proper basic imaging formation algorithm for the high efficiency of the Stolt interpolation and the low sensitivity to squint angle. It should be highlighted that one significant defect is that Stolt interpolation would modify the characteristics of motion errors and bring about extra nonnegligible range-dependent MTRC in addition to APEs. Theoretical derivation indicates that, for the RMA, the most effective approach is to compensate for the original motion errors before the Stolt interpolation. Therefore, motion estimation can be regarded as one of the most significant issues in UAV SAR imaging.

In this article, we proposed a robust estimation algorithm purely based on symmetric triangle modulated FMCW signal interferometry to retrieve original motion errors during data collection. In the implementation of our method, the high-order motion errors can be easily obtained, and the critical concern is to eliminate the normal phase history, which mostly consists of a linear and a quadric component determined by the distribution of the observed target. With this preknowledge of the structural composition of the interferometric phase, the second-order error is estimated by APWM, and the linear element is eliminated by gradient matching and joining. Simulations validate the correctness of the built signal

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model. Moreover, the results also demonstrate that our proposed method achieves the best performance with prominent points, while an acceptable accuracy at the subwavelength level with high probability relies on linear targets that have strong and distinguishable scattering characteristics and are relatively isolated in the azimuth dimension. Furthermore, the presented estimation algorithm is applied to the data acquired in the real ground experiments with a W-band UAV SAR system. For one thing, the comparisons of motion parameters between IMU/GPS measured and our estimated values suggest that the proposed method can achieve higher accuracy. That is, our proposed method can be used to analyze and correct the measurement errors of traditional navigation systems in depth. More importantly, it also confirms the ability of fine imaging without auxiliary data from external motion measurement sensors, from which we can greatly benefit in terms of the reduction of size, weight, and cost for system hardware implementation. Finally, we provide a comparison between our proposed method and two other postprocessing algorithms. The results demonstrate that the proposed strategy has a lower requirement for the scene and processor in UAV SAR MoCo.

Nevertheless, the analysis and correction of motion errors in this article mainly aim at strip-map mode SAR signals, and the proposed method cannot be directly applied to the spotlight mode. However, we may benefit from some unique characteristics of the spot-mode signal in the azimuth frequency domain. A robust and feasible autofocus approach that, based on the STLFMCW signal, requires more attention to detail, and we will examine this in the future.

Appendix A

To obtain the expression of $\phi_e(K_x^*, \Delta K_y)$, we applied first-order Taylor series expansion at $\Delta K_y = 0$

$$\begin{aligned}\phi_e\left(\Delta K_y, K_x^*\right)\big|_{\Delta K_y=0} &= \sqrt{\left(\Delta K_y \sec\theta + \sqrt{K_{rc}^2 - K_x^2}\right)^2 + K_x^2}\Delta R\left(X^*\right)\Big|_{\Delta K_y=0} \\ &= -K_{rc}\Delta R\left(X^*\right) \qquad \text{(A1)}\end{aligned}$$

$$\begin{aligned}&\left.\frac{\partial\phi_e\left(K_x^*, \Delta K_y\right)}{\partial \Delta K_y}\right|_{\Delta K_y=0} \\ &= -\left(\frac{\partial K_r}{\partial \Delta K_y}\Delta R\left(X^*\right) + K_r\frac{\partial \Delta R(X^*)}{\partial \Delta K_y}\right)\Big|_{\Delta K_y=0}. \qquad \text{(A2)}\end{aligned}$$

To be brief, let θ' be the instantaneous squint angle, and

$$\sin\theta' = \frac{X - r\sin\theta - x}{\sqrt{(r\cos\theta)^2 + (X - r\sin\theta - x)^2}}.$$

Then, we can obtain that

$$\begin{aligned}\left.\frac{\partial \Delta R\left(X^*\left(K_x^*\right)\right)}{\partial \Delta K_y}\right|_{\Delta K_y=0} &= \left.\frac{\partial \Delta R(X)}{\partial X}\right|_{X=X^*\left(K_x^*\right)} \cdot \frac{\partial X^*\left(K_x^*\right)}{\partial K_r}\cdot\frac{\partial K_r}{\partial \Delta K_y} \\ &= \frac{\partial \Delta R(X)}{\partial X}\frac{r\tan\theta'}{K_{rc}\cos\theta'} \qquad \text{(A3)}\end{aligned}$$

with

$$\left.\frac{\partial \Delta R(X)}{\partial X}\right|_{X=X^*\left(K_x^*\right)} = \frac{\partial \Delta R(X)}{\partial X} \qquad \text{(A3.1)}$$

$$\begin{aligned}\left.\frac{\partial X^*\left(K_x^*\right)}{\partial K_r}\right|_{\Delta K_y=0} &= \left.\frac{r\cos\theta K_x^* K_r}{\left(K_r^2 - K_x^{*2}\right)^{\frac{3}{2}}}\right|_{\Delta K_y=0} \\ &= \frac{r\cos\theta\sin\theta'}{K_{rc}\cos^3\theta'} \qquad \text{(A3.2)}\end{aligned}$$

$$\begin{aligned}\left.\frac{\partial K_r}{\partial \Delta K_y}\right|_{\Delta K_y=0} &= \left.\frac{\frac{\Delta K_y}{\cos^2\theta} + \frac{\sqrt{K_{rc}^2 + K_x^{*2}}}{\cos\theta}}{\sqrt{\left(\frac{\Delta K_y}{\cos\theta} + \sqrt{K_{rc}^2 - K_x^{*2}}\right)^2 + K_x^{*2}}}\right|_{\Delta K_y=0} \\ &= \frac{\sqrt{K_{rc}^2 + K_x^{*2}}}{K_{rc}\cos\theta} = \frac{\cos\theta'}{\cos\theta}. \qquad \text{(A3.3)}\end{aligned}$$

Finally, the phase error can be expressed as

$$\phi_e\left(X, \Delta K_y\right) = \zeta_0(X) + \Delta K_y\zeta_1(X) \qquad \text{(A4)}$$

where

$$\zeta_0(X) = -K_{rc}\Delta R(X) \qquad \text{(A4.1)}$$

$$\zeta_1(X) = -\frac{\cos\theta'}{\cos\theta}\Delta R(X) - \frac{r\tan\theta'}{\cos\theta'}\frac{\partial \Delta R(X)}{\partial X}. \qquad \text{(A4.2)}$$

Appendix B

To perform azimuth FT on (21), POSP is ready to be used. First, we get the phase function, i.e.,

$$\Phi\left(K_y, K_x\right) = -j2K_{rc}\sqrt{(r\cos\theta)^2 + (X - x_0 - r\sin\theta)^2} - jK_xX. \qquad \text{(B1)}$$

Equating the first derivative of B1 with respect to X with 0 yields

$$K_{rc}\frac{\partial\sqrt{(r\cos\theta)^2 + (X - x_0 - r\sin\theta)^2}}{\partial X} + K_x = 0. \qquad \text{(B2)}$$

The solution of the above equation is the SP point, i.e.,

$$X^* = -\frac{K_x r\cos\theta}{\sqrt{4K_{rc}^2 + K_x^2}} + x_0 + r\sin\theta. \qquad \text{(B3)}$$

Then, we substitute X^* for X in (B1). The phase of reference spectrum is given by

$$\Phi\left(K_y, K_x\right) = -r\cos\theta\sqrt{(2K_{rc})^2 - K_x^2} - K_x(r\sin\theta + x_0). \qquad \text{(B4)}$$

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